

Timestamp	Comments
	During the Thesis 1 presentation, it was noted that if a single feature determines whether an instance is an attack, it could invalidate the clustering process. If only one indicator exists—for example, if one feature out of sixteen explicitly identifies an attack (1 = attack, 0 = no attack)—then the remaining fifteen features become irrelevant. In such a case, machine learning would be unnecessary.
	Make sure the dataset is not dependent on a single feature to determine attack or not
2:40 - 4:50	Regarding the attacks: do you have variations, such as different parameters in the scripts, or are the runs always identical? While the number of nodes is constant, have you introduced any randomness into the packet intervals or other metrics, aside from the topology? Although this is currently for Thesis 2, for Thesis 3 you will need more than just hardcoded scripts; the current data is too limited because the behavior remains the same across multiple runs, resulting in identical patterns for each attack.
5:15 - 7:10	Regarding the 16 out of 24 runs: have you balanced the dataset across these test runs? This represents only one variation in topology. Since the data is not visible, it is unclear if the topology has any effect; despite the variations in attack (e.g., four black holes), the behavior remains identical regardless of the topology.
8:40- 8:55	Do the runs in public places and different variations aside from house only
9:10-12:00	Do you have a specific testbed scenario, such as smart farming? Currently, the node deployment and subsequent attacks appear random. To improve this, you could reference existing papers to define a realistic deployment scenario and topology. Clarify the IoT context—for example, is it an agriculture setting with constant data transmission, or a smart home with random traffic? It is important to specify the application context for the mesh network.
12:45-16:00	If there is no difference in running r1 till r3 it is waste if it is the same output add variance for each runs . variation is like how the attacks will be different from each run like putting different position of the attackers . find a paper that talks about target deployment Elaborate: If there is no distinction between experimental runs (from r1 through r3), executing multiple iterations yields redundant outputs and represents an inefficient use of resources. To maximize the utility of the dataset, it is essential to introduce controlled variance across individual runs. This variation can be achieved by modifying the specific parameters of each execution, such as altering the physical or logical positions of the attacker nodes within the mesh network topology. By changing the spatial deployment of adversarial elements during each run, the dataset can capture a broader spectrum of localized behavioral patterns and their varying effects on adjacent network nodes. To ensure these modifications reflect realistic network environments, existing literature on target deployment and node arrangement strategies should be referenced to establish structured and scientifically justified topologies.

17:50-18:30	Is there a way you can verify if the attack is correct where did your script based or reference cus if it can do in topology but the attack is wrong the data is wrong
44:30-49:00	Elaborate: It is necessary to establish a clear baseline for "benign" traffic by characterizing normal network behavior under various conditions. The current clustering results appear biased because the scenarios for both benign and malicious traffic are identical, lacking sufficient variation. To address this, you should introduce realistic topologies and a broader range of network conditions. This will help differentiate between instances of high network utilization—where traffic load is naturally high but legitimate—and actual malicious flooding, ensuring the model can accurately distinguish between the two based on behavioral patterns rather than just traffic volume. Show how you validated the attacks; if the scripts are based on existing literature, please provide your basis. Even if created from scratch, do you have documentation or an online database that details the steps of these attacks to verify their accuracy? Regarding the 10k datasets, while you can continue using your current setup, you do not necessarily need to run each test for a full hour. Instead, consider running smaller intervals with more variations; this approach will allow you to collect a wider range of test cases.